A quantitative central limit theorem for tangential random fields on stratified spaces*

Victor Amaya† Nicholas Cook‡ Jonathan Mattingly§

Abstract

We establish a quantitative central limit theorem for tangential random fields on smoothly stratified $\mathrm{CAT}(\kappa)$ spaces, where the population barycenter may be a point at which the tangent space is a metric cone and not a vector space. Following the construction of Mattingly, Miller and Tran, sample points are encoded as a centered real-valued random field τ indexed by the unit tangent cone $\mathcal{S}_{\bar{\mu}}\mathcal{M}$, and we quantify the known qualitative central limit theorem for the standardized empirical field $\bar{\tau}_n = n^{-1/2}\sum_{i=1}^n \tau_i$. Under a third-moment condition on τ we develop a version of Stein's method on the Hilbert space $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$, built from the spectral decomposition of the covariance operator and a Hilbert–Gaussian integration-by-parts formula, which gives the rate $\mathcal{O}(n^{-1/2})$ in a smooth metric. Assuming in addition that the eigenvalues of the covariance operator decay polynomially, $\lambda_k = \mathcal{O}(k^{-2s})$ for some $s > 1$, and that its eigenfunctions are uniformly bounded, a Gaussian smoothing argument adapted from Balasubramanian, Goldstein, Ross and Salim transfers the estimate to the 1-Wasserstein distance with respect to the supremum norm on $C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$, with rate

$$d_{\mathcal{W}}(\bar{\tau}_n, G) = \mathcal{O}\Big(n^{-\frac{s-1}{6s-1}}\Big),$$

where G is the limiting Gaussian tangent field. We illustrate the theory on the *kale*, a planar cone with an isolated hyperbolic singularity; there $s = 3$ and the rate is $\mathcal{O}(n^{-2/17})$.

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*In memory of Sayan Mukherjee

†Duke University, Durham, USA. E-mail: vamayacar@gmail.com
‡Duke University, Durham, USA. E-mail: author@institution.edu
§Duke University, Durham, USA. E-mail: author@institution.edu

1 Introduction

Central limit theorems have been studied extensively on smooth manifolds. A difficulty intrinsic to nonlinear spaces is the absence of vector addition, which prevents a translation of the classical Euclidean statement to the manifold setting. The standard approach is to map the sample points to the tangent space $T_{\bar{\mu}}\mathcal{M}$ at a (strategically chosen) reference point $\bar{\mu}$ and to prove a limit theorem there. In several statistical applications, however, force the point $\bar{\mu}$ to be a *singular* one of the underlying space [12, 14]. At such a point the tangent space $T_{\bar{\mu}}\mathcal{M}$ is no longer a vector space but a metric *cone* instead. A qualitatively new limiting phenomena appear on $T_{\bar{\mu}}\mathcal{M}$, the limit may be "sticky", or it may be a nonstandard Gaussian object supported on the cone. These effects were first analyzed for open books and for isolated planar singularities by Hotz, Huckemann, Mattingly, Miller, Nolen and coauthors [8, 9], and for Fréchet means on phylogenetic tree space by Barden, Le and Owen [4].

To accommodate a conical tangent space, Mattingly, Miller and Tran [14] encode each sample point as a real-valued *tangential random field* indexed by the unit tangent cone $\mathcal{S}_{\bar{\mu}}\mathcal{M} \subseteq T_{\bar{\mu}}\mathcal{M}$, also called the *set of directions*. Concretely, let $x_1, \dots, x_n$ be independent and identically distributed (i.i.d.) samples from a probability measure μ on a smoothly stratified space $\mathcal{M}$ with population barycenter $\bar{\mu}$, and let $\log_{\bar{\mu}}$ denote the logarithm map onto $T_{\bar{\mu}}\mathcal{M}$. The induced tangent field of a sample $x \sim \mu$ is the centered field

$$\tau(x, V) = \langle \log_{\bar{\mu}} x,\, V\rangle_{\bar{\mu}} - \mathbb{E}_{x\sim\mu}\langle \log_{\bar{\mu}} x,\, V\rangle_{\bar{\mu}}, \qquad V \in \mathcal{S}_{\bar{\mu}}\mathcal{M}, \tag{1.1}$$

where $\langle \cdot, \cdot \rangle_{\bar{\mu}}$ is the angular pairing on the cone (made precise in Section 2). Writing $\tau_i = \tau(x_i, \cdot)$ for the field associated to the i-th sample, [14, Theorem 2] shows that the *standardized empirical field*

$$\bar{\tau}_n(V) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \tau_i(V), \qquad V \in \mathcal{S}_{\bar{\mu}}\mathcal{M}, \tag{1.2}$$

converges in distribution, as a random element of $(C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}), \|\cdot\|_\infty)$, to a centered Gaussian tangent field G with covariance function $R(U, V) = \mathbb{E}_\mu[\tau(x, U)\tau(x, V)]$. That result is qualitative, that is, it does not quantify how fast the convergence takes place.

1.1 Main result

In this paper is to provide quantitative study of the problem previously described. We measure the discrepancy between $\bar{\tau}_n$ and G in the 1-Wasserstein distance $d_{\mathcal{W}}$ associated with the supremum norm on $C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$. Our main result is the following rate of convergence; the relevant objects are defined in Section 2 and the proof is given in Section 7.

Theorem 1.1. *Let μ be a localized measure on a smoothly stratified $\mathrm{CAT}(\kappa)$ space $\mathcal{M}$, and let G be the Gaussian tangent field induced by μ, with continuous covariance function R. Let $\bar{\tau}_n = n^{-1/2}\sum_{i=1}^n \tau_i$ be the standardized empirical field of a collection of i.i.d. centered induced tangent fields τ_i with covariance R, satisfying the third-moment condition $\mathbb{E}\|\tau\|^3_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} < \infty$. Assume the spectral decomposition of the covariance operator T_R satisfies that its eigenvalues decay as $\lambda_k = \mathcal{O}(k^{-2s})$ for some $s > 1$ and its eigenfunctions are uniformly bounded (in the precise sense of Assumption 7.1). Then*

$$d_{\mathcal{W}}(\bar{\tau}_n, G) = \mathcal{O}\Big(n^{-\frac{s-1}{6s-1}}\Big). \tag{1.3}$$

In particular, the right-hand side of (1.3) tends to zero as $n \to \infty$, so Theorem 1.1 recovers, and makes quantitative, the qualitative convergence $\bar{\tau}_n \xrightarrow{w} G$ of [14].

Remark 1.2. The exponent in (1.3) is one point on a curve. Writing

$$\alpha(s) = \frac{s-1}{6s-1}, \qquad s > 1,$$

the proof balances the three terms of (7.9) at the decay exponent s of Assumption 7.1 and yields $d_{\mathcal{W}}(\bar{\tau}_n, G) = \mathcal{O}(n^{-\alpha(s)})$. The map α is strictly increasing on $(1, \infty)$ with $\alpha(s) \uparrow 1/6$ as $s \to \infty$, so $1/6$ is a supremum attained by no finite s.

Corollary 1.3. *Under the hypotheses of Theorem 1.1, $\bar{\tau}_n$ converges weakly to G in $(C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}), \|\cdot\|_\infty)$.*

1.2 Method of proof

The proof proceeds in two stages, each corresponding to one integral-probability metric. Recall that for a class $\mathcal{H}$ of real-valued test functionals on $C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$, the induced distance between two random fields F, H is

$$d_{\mathcal{H}}(F, H) = \sup_{\zeta \in \mathcal{H}} \big|\mathbb{E}[\zeta(F)] - \mathbb{E}[\zeta(H)]\big|. \tag{1.4}$$

The Wasserstein distance $d_{\mathcal{W}}$ corresponds to the class $\mathcal{W}$ of 1-Lipschitz functionals with respect to $\|\cdot\|_\infty$. We first work with a smaller, smoother class $\mathcal{F}$ of functionals with uniformly bounded Fréchet derivatives up to order three (Definition 2.10), which yields a fast rate, and then transfer the bound from $d_{\mathcal{F}}$ to $d_{\mathcal{W}}$.

Act one. A smooth metric (Sections 3–6). We develop a version of Stein's method on the Hilbert space $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$. The guiding idea is taken from the multivariate Gaussian case of Meckes [15], rewriting the multivariate Stein operator in the eigenbasis of the covariance matrix exposes a form that generalizes to infinite dimensions once the eigenbasis is replaced by the eigenfunctions $\{\psi_k\}_k$ of the covariance operator T_R. This generalization leads us to the (infinite-dimensional) Stein equation

$$(\Delta\Lambda - f \cdot \nabla\Lambda)(f) = \Gamma(f) - \mathbb{E}\,\Gamma(G), \qquad f \in C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}), \tag{1.5}$$

for test functionals $\Gamma\colon C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}) \to \mathbb{R}$, where Δ and ∇ are defined spectrally through T_R (Definition 4.1). We show that (1.5) admits the explicit solution

$$\Lambda(f) = \int_0^1 \frac{1}{2t}\Big[\,\mathbb{E}\,\Gamma\big(\sqrt{t}\,f + \sqrt{1-t}\,G\big) - \mathbb{E}\,\Gamma(G)\,\Big]\,dt, \tag{1.6}$$

the infinite-dimensional analogue of the Meckes solution. The verification relies on a Hilbert–Gaussian integration-by-parts identity tailored to tangential Gaussian fields (Proposition 5.2) which states that for $X \sim \gamma$, a centered Gaussian random element of $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ with covariance operator T_R, and F Fréchet differentiable with bounded derivative, the following holds

$$\mathbb{E}\big(\langle X, \psi_k\rangle_{L^2}\, F(X)\big) = \lambda_k\, \mathbb{E}\big(DF(X)[\psi_k]\big). \tag{1.7}$$

A Taylor expansion of Λ along the leave-one-out fields $\bar{\tau}_n^{(i)}$, combined with (1.7) and the trace-class property of T_R, controls the resulting error terms and gives, for all $\Gamma \in \mathcal{F}$,

$$d_{\mathcal{F}}(\bar{\tau}_n, G) = \mathcal{O}(n^{-1/2}). \tag{1.8}$$

Act two. From $d_{\mathcal{F}}$ to Wasserstein (Section 7). Is not clear if the smooth metric $d_{\mathcal{F}}$ is strong enough to imply tightness (see [1, Section 1.1]). To upgrade (1.8) to the Wasserstein distance we adapt the Gaussian smoothing technique of Balasubramanian,

Goldstein, Ross and Salim [1]. A 1-Lipschitz functional ζ is replaced by a smoothed functional $\zeta_{t,\delta}(f) = \mathbb{E}\,\zeta(P_t f + \delta S_t)$ with three bounded derivatives, whose constants blow up as $t, \delta \to 0$, and one pays for the replacement in the supremum norm. Two features distinguish our implementation from [1], whose analysis was developed for fields indexed by the Euclidean sphere. First, the regularization P_t is the spectral truncation of the covariance operator itself (only the eigenmodes with $\lambda_k \geq t$ are kept), instead of using the heat semigroup of the index set, we use a smoothing Gaussian S_t white noise on the retained modes. Thus obtaining a classical Gaussian mollification in some finite $\mathbb{R}^N$. Second, every quantity entering the smoothing inequality is controlled by an elementary estimate on the spectrum of T_R. This is where the eigenvalue decay and the eigenfunction bounds of Assumption 7.1 enter, and it is the reason why no geometric input on $\mathcal{S}_{\bar{\mu}}\mathcal{M}$ beyond its compactness is required. Optimizing t and δ against (1.8) gives the rate of Theorem 1.1. The slower rate in comparison with the clasical Central Limit Theorem rate of $n^{-1/2}$ in Euclidean spaces is the price of passing from the smooth metric to the Wasserstein metric.

1.3 Related work

Stein's method for Gaussian and functional approximation goes back to Barbour [2] and has been developed in many directions; the recent work of Barbour, Ross and Zheng [3] treats smoothing and functional approximation, and Balasubramanian, Goldstein, Ross and Salim [1] obtain Wasserstein bounds for Gaussian random field approximation with applications to wide neural networks. The multivariate Gaussian case underlying our spectral reformulation is due to Meckes [15]. The geometric and probabilistic framework of stratified spaces, localized measures and induced tangent fields is from Mattingly, Miller and Tran [12, 13, 14], whose qualitative central limit theorem [14, Theorem 2] we make quantitative. Our infinite-dimensional integration-by-parts formula is in the spirit of the Cameron–Martin theory for Gaussian measures on Hilbert spaces [6, 11].

1.4 Organization

Section 2 fixes the geometric and probabilistic setting and the function classes used throughout. Section 3 reviews Stein's method in finite dimensions and recasts it using the eigenbasis of the covariance matrix. Section 4 introduces the infinite-dimensional Stein equation and its candidate solution. Section 5 proves the Hilbert–Gaussian integration-by-parts formula and verifies that the candidate solves the Stein equation. Section 6 establishes the $\mathcal{O}(n^{-1/2})$ rate in the smooth metric. Section 7 carries out the Gaussian smoothing and proves Theorem 1.1. Finally, Section 8 works out the kale example explicitly.

2 Setting and preliminaries

Throughout, $(\mathcal{M}, \mathbf{d})$ is a smoothly stratified metric space that is $\mathrm{CAT}(\kappa)$ for some κ, and μ is a localized Borel probability measure on $\mathcal{M}$ in the sense of [12], with $\mathbb{E}_\mu\, \mathbf{d}(\bar{\mu}, x)^2 < \infty$. We assume that μ has a unique Fréchet mean (barycenter) $\bar{\mu} \in \mathcal{M}$,

$$\bar{\mu} \coloneqq \operatorname*{arg\,min}_{p \in \mathcal{M}} \mathbb{E}_{x \sim \mu}\, \mathbf{d}(p, x)^2,$$

which is allowed to be a singular point of $\mathcal{M}$. We recall the objects from [12, 14] that are needed below; the reader is referred to those works for the full construction.

2.1 The tangent cone and the set of directions

At the barycenter $\bar{\mu}$ the space $\mathcal{M}$ possesses a tangent cone $T_{\bar{\mu}}\mathcal{M}$, a metric cone with apex $\mathbf{0}$ which, when $\bar{\mu}$ is singular, is not a vector space. The *logarithm map* $\log_{\bar{\mu}}\colon \mathcal{M} \to T_{\bar{\mu}}\mathcal{M}$ sends a point x to the initial velocity of the unit-speed geodesic from $\bar{\mu}$ to x, scaled by $\mathbf{d}(\bar{\mu}, x)$. The *unit tangent cone*, or *set of directions*, is

$$\mathcal{S}_{\bar{\mu}}\mathcal{M} \coloneqq \{\, V \in T_{\bar{\mu}}\mathcal{M} : \|V\| = 1 \,\},$$

a compact metric space when equipped with the angular metric.

Remark 2.1. Since $\mathcal{S}_{\bar{\mu}}\mathcal{M}$ is compact, the space $C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ of continuous real-valued functions on $\mathcal{S}_{\bar{\mu}}\mathcal{M}$, equipped with the supremum norm $\|\cdot\|_\infty$, is a separable Banach space, and $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ (with respect to a fixed finite, strictly positive Borel measure on $\mathcal{S}_{\bar{\mu}}\mathcal{M}$) is a separable Hilbert space. We write $\langle \cdot, \cdot \rangle_{L^2}$ for its inner product.

Definition 2.2 (Angular pairing)**.** *For $x \in \mathcal{M}$ and $V \in \mathcal{S}_{\bar{\mu}}\mathcal{M}$ the* angular pairing *is*

$$\langle \log_{\bar{\mu}} x,\, V \rangle_{\bar{\mu}} = \mathbf{d}(\bar{\mu}, x)\, \cos\big(\angle(\log_{\bar{\mu}} x,\, V)\big),$$

where $\angle(\cdot, \cdot)$ denotes the angle at the apex of the cone.

2.2 Induced tangent fields and the Gaussian limit

Definition 2.3 (Induced tangent field)**.** *The* tangent field induced by *a random point $x \sim \mu$ is the centered real-valued random field on $\mathcal{S}_{\bar{\mu}}\mathcal{M}$ given by* (1.1)*. For i.i.d. samples $x_1, \dots, x_n \sim \mu$ we write $\tau_i = \tau(x_i, \cdot)$ and regard each τ_i as a random element of $C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$; by [14, Prop. 3.2] the fields τ_i have a.s. Lipschitz-continuous sample paths whenever $\mathbb{E}_\mu\, \mathbf{d}(\bar{\mu}, x) < \infty$. To lighten the notation we suppress the dependence on μ and write $\tau(V)$ for $\tau(\mu, V)$, and similarly for objects derived from μ.*

Definition 2.4 (Covariance function and covariance operator)**.** *The* covariance function *of μ is*

$$R(V, W) \coloneqq \mathbb{E}_\mu\big[\tau(x, V)\, \tau(x, W)\big], \qquad V, W \in \mathcal{S}_{\bar{\mu}}\mathcal{M},$$

a continuous, symmetric, positive-semidefinite kernel.

We define as well its covariance operator $T_R\colon L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}) \to L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ *as*

$$T_R f(V) \coloneqq \int_{\mathcal{S}_{\bar{\mu}}\mathcal{M}} R(V, W)\, f(W)\, dW.$$

By Mercer's theorem [7] the operator T_R is compact, self-adjoint and positive, with eigenvalues $\lambda_k \geq 0$, $k \geq 1$ (repeated according to multiplicity and accumulating only at 0), and an orthonormal basis $\{\psi_k\}_{k \geq 1}$ of $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ of eigenfunctions, $T_R\psi_k = \lambda_k \psi_k$, and $R(V, W) = \sum_{k \geq 1} \lambda_k \psi_k(V) \psi_k(W)$, where the sum converges absolutely and uniformly. Each positive eigenvalue has finite multiplicity, so the positive eigenvalues can be listed in nonincreasing order, repeated according to multiplicity; we write

$$\lambda_{(1)} \geq \lambda_{(2)} \geq \cdots > 0$$

for this enumeration (a finite list if T_R has finite rank). The index k of the full eigenbasis, in contrast, is not assumed to order the eigenvalues.

We write $\operatorname{Tr}(T_R) = \sum_{k \geq 1} \lambda_k$ for the *trace* of the covariance operator.

Lemma 2.5 (Trace class)**.** *The covariance operator T_R is trace class, i.e. $\sum_{k \geq 1} \lambda_k < \infty$.*

Proof. Since R is continuous on the compact set $\mathcal{S}_{\bar{\mu}}\mathcal{M} \times \mathcal{S}_{\bar{\mu}}\mathcal{M}$, Mercer's theorem gives $\operatorname{Tr}(T_R) = \sum_{k \geq 1} \lambda_k = \int_{\mathcal{S}_{\bar{\mu}}\mathcal{M}} R(V, V)\, dV < \infty$. □

Definition 2.6 (Gaussian tangent field)**.** *The* Gaussian tangent field induced by μ *is the centered Gaussian random field G on $\mathcal{S}_{\bar{\mu}}\mathcal{M}$ with covariance function R. By the Karhunen–Loève expansion,*

$$G = \sum_{k\geq 1} \sqrt{\lambda_k}\, \eta_k\, \psi_k, \qquad \eta_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1), \tag{2.1}$$

with convergence in L^2 of the partial sums and a.s. in $C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$. We denote by γ the law of G, a centered Gaussian measure on $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$, and write $G_k = \langle G,\psi_k\rangle_{L^2} = \sqrt{\lambda_k}\,\eta_k$.

We record the second-order structure of the coordinates of τ and G in the eigenbasis, which is used repeatedly.

Lemma 2.7. *For each centered induced tangent field τ,*

$$\mathbb{E}\,\langle \tau,\psi_k\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M})} = 0, \qquad \mathbb{E}\big[\langle \tau,\psi_k\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M})}\langle \tau,\psi_j\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M})}\big] = \delta_{kj}\,\lambda_k.$$

Proof. The first identity holds because τ is centered. For the second, by Fubini's theorem and the definition of R,

$$\begin{aligned}\mathbb{E}\big[\langle \tau,\psi_k\rangle_{L^2}\langle \tau,\psi_j\rangle_{L^2}\big] &= \int_{\mathcal{S}_{\bar{\mu}}\mathcal{M}}\int_{\mathcal{S}_{\bar{\mu}}\mathcal{M}} R(V,W)\,\psi_k(V)\psi_j(W)\,dV\,dW \\ &= \langle T_R\psi_k,\psi_j\rangle_{L^2} = \delta_{kj}\lambda_k.\end{aligned}$$

□

Remark 2.8 (Ordering of the spectrum and zero eigenvalues)**.** The kernel of T_R may be nontrivial, and even infinite-dimensional (this happens for the kale, Section 8). If $\ker T_R \neq \{0\}$ and T_R has infinitely many positive eigenvalues, then no enumeration of the full eigenbasis $\{\psi_k\}_{k\geq 1}$ over $\mathbb{N}$ makes $k\mapsto\lambda_k$ nonincreasing, because every zero eigenvalue would have to be preceded by infinitely many positive ones. For this reason k denotes an arbitrary, fixed enumeration of the eigenbasis, and whenever the order of the eigenvalues matters (Assumption 7.1) we use the nonincreasing enumeration $(\lambda_{(j)})_j$ of the *positive* eigenvalues. When $\ker T_R = \{0\}$ one may simply take $\lambda_k = \lambda_{(k)}$.

2.3 The Cameron–Martin space

The Cameron–Martin space of the Gaussian measure γ is $H(R) = T_R^{1/2}\big(L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})\big)$, with inner product

$$\langle f,g\rangle_H = \langle T_R^{-1/2}f,\, T_R^{-1/2}g\rangle_{L^2} = \sum_{k\geq 1}\frac{\langle f,\psi_k\rangle_{L^2}\,\langle g,\psi_k\rangle_{L^2}}{\lambda_k},$$

the sum being over those k with $\lambda_k > 0$ (see [6, 11]). Each eigenfunction ψ_k with $\lambda_k>0$ belongs to $H = H(R)$. The Cameron–Martin theorem will be recalled in the form we need in Section 5.

2.4 Fréchet derivatives and test functionals

We use the Fréchet calculus on $C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$. For a functional $F\colon C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})\to\mathbb{R}$ and $f\in C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$, the j-th Fréchet derivative $D^jF(f)$ is the bounded symmetric j-linear form on $C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$ obtained by differentiating F at f. Unless stated otherwise, *all Fréchet derivatives are taken with respect to the $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$ norm on the input space $C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$ and the Euclidean norm on $\mathbb{R}$*; for brevity we then simply write "Fréchet differentiable".

Definition 2.9 (Operator norm of a k-linear form)**.** *For a k-linear form $A\colon \big(C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})\big)^k\to\mathbb{R}$ we set*

$$\|A\|_{\mathrm{op}} = \sup_{\|f\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M})}=1} \big|A[f,\dots,f]\big|.$$

By convention the operator norm of a k-linear form always refers to the $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$ norm on the input space. We also write $\|DF\|_{\mathrm{op}} = \sup_g \|DF(g)\|_{\mathrm{op}}$.

Definition 2.10 (Smooth test functionals)**.** *The class of* smooth test functionals *is*

$$\mathcal{F} = \Big\{ \zeta\colon C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R}) \to \mathbb{R} : \sup_g \|D^k\zeta(g)\|_{\mathrm{op}} \le 1,\ k=1,2,3 \Big\}. \tag{2.2}$$

Definition 2.11 (Wasserstein class)**.** *The* Wasserstein class *is the set of functionals that are 1-Lipschitz with respect to the supremum norm,*

$$\mathcal{W} \coloneqq \Big\{ \zeta\colon C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R}) \to \mathbb{R} \,:\, \sup_{f\neq g} \frac{|\zeta(f)-\zeta(g)|}{\|f-g\|_\infty} \le 1 \Big\}.$$

The induced distance $d_{\mathcal{W}}$ (see Equation (1.4)*) is the 1-Wasserstein distance.*

Let us recall that convergence in $d_{\mathcal{W}}$ implies weak convergence on the Polish space $(C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R}),\|\cdot\|_\infty)$. Moreover, on an unbounded metric space such as this one, $d_{\mathcal{W}}$ metrizes weak convergence together with convergence of first moments, on the set of laws with finite first moment.

Asymptotic notation. For real quantities f, g depending on one or more parameters we write $f = \mathcal{O}(g)$ and $f \lesssim g$ to mean $|f| \le Cg$ for a constant $C > 0$ that may depend on the fixed parameters of the problem (such as the moments of τ, and the spectrum of T_R) but not on n.

3 Stein's method in finite dimensions

We briefly review the finite-dimensional Gaussian Stein's method and then recast it in the eigenbasis of the covariance. The reformulation is the bridge to the infinite-dimensional setting of Section 4: it is invariant under the choice of dimension and depends only on the spectral data of the covariance.

3.1 The multivariate Stein operator

The starting point is the following characterization of the multivariate Gaussian, due to Meckes [15, Lemma 1]. Let $Z \in \mathbb{R}^d$ have i.i.d. standard Gaussian coordinates and let $Z_\Sigma = \Sigma^{1/2}Z$ for a symmetric positive-semidefinite covariance matrix Σ. Then for every twice continuously differentiable, compactly supported $f\colon \mathbb{R}^d \to \mathbb{R}$,

$$\mathbb{E}\Big[\, \langle Z_\Sigma, \nabla f(Z_\Sigma)\rangle - \langle \mathrm{Hess}\, f(Z_\Sigma), \Sigma\rangle_{H.S.} \,\Big] = 0, \tag{3.1}$$

where $\langle\cdot,\cdot\rangle_{H.S.}$ is the Hilbert–Schmidt inner product; moreover this identity, holding for all such f, characterizes the law of Z_Σ. Finally, for $g \in C^\infty(\mathbb{R}^d)$ the function

$$U_o g(x) = \int_0^1 \frac{1}{2t}\big[\, \mathbb{E}\, g(\sqrt{t}\,x + \sqrt{1-t}\,Z_\Sigma) - \mathbb{E}\, g(Z_\Sigma)\,\big]\, dt \tag{3.2}$$

solves the Stein equation $\langle x, \nabla h(x)\rangle - \langle \mathrm{Hess}\, h(x), \Sigma\rangle_{H.S.} = g(x) - \mathbb{E}\, g(Z_\Sigma)$.

Writing the integrand of (3.1) as an operator,

$$Af(x) = \langle x, \nabla f(x)\rangle - \mathrm{Tr}\big(\Sigma^{1/2}\,\nabla^2 f(x)\,\Sigma^{1/2}\big) = \sum_{i=1}^d \frac{\partial f(x)}{\partial x_i}\, x_i - \sum_{i,j=1}^d \frac{\partial^2 f(x)}{\partial x_i \partial x_j}\,\Sigma_{ji}. \tag{3.3}$$

Remark 3.1. When $d = 1$ one has $\Sigma^{1/2} = \sigma$ and $Af(x) = \sigma^2 f''(x) - xf'(x)$, the classical one-dimensional Stein operator. The second-order term in higher dimensions encodes the correlation structure of the coordinates.

3.2 The Stein operator in the covariance eigenbasis

We now diagonalize (3.3). Let $\{(\lambda_k, v_k)\}_{k=1}^d$ be the eigenpairs of Σ, so that $\Sigma = \sum_k \lambda_k v_k v_k^\top$, and write $D_v h$ for the directional derivative of h along v.

Lemma 3.2. *With the notation above, the multivariate Stein operator can be written*

$$Ah(x) = \sum_{i=1}^{d} \langle x, v_i \rangle \, D_{v_i} h(x) - \sum_{i=1}^{d} \lambda_i \, D^2_{v_i v_i} h(x). \tag{3.4}$$

Proof. For the first-order term, expanding x and $\nabla h(x)$ in the orthonormal eigenbasis,

$$\langle x, \nabla h(x) \rangle = \Big\langle \sum_i \langle x, v_i \rangle v_i, \; \sum_i (D_{v_i} h(x)) v_i \Big\rangle = \sum_{i=1}^{d} \langle x, v_i \rangle \, D_{v_i} h(x).$$

For the second-order term, using $D_{v_j} D_{v_i} h(x) = v_j^\top \nabla^2 h(x) v_i$ and $\operatorname{Tr}(AB) = \sum_{i,j} A_{ij} B_{ji}$,

$$\operatorname{Tr}\big(\Sigma \, \nabla^2 h(x)\big) = \operatorname{Tr}\Big(\big(\textstyle\sum_k \lambda_k v_k v_k^\top\big) \nabla^2 h(x) \Big) = \textstyle\sum_{i=1}^d \lambda_i \, D^2_{v_i v_i} h(x).$$

Substituting both into (3.3) gives (3.4). □

The solution (3.2) admits a parallel representation. Using the rotational invariance of the standard Gaussian to write $\Sigma^{1/2} Z = \sum_k \lambda_k^{1/2} \eta_k v_k$ with η_k i.i.d. standard Gaussian, one obtains

$$U_o g(x) = \int_0^1 \frac{1}{2t} \Big[\mathbb{E}\, g\Big(\textstyle\sum_k \sqrt{t}\, \langle x, v_k \rangle v_k + \sqrt{1-t}\, \lambda_k^{1/2} \eta_k v_k \Big) - \mathbb{E}\, g\Big(\textstyle\sum_k \lambda_k^{1/2} \eta_k v_k \Big) \Big] \, dt. \tag{3.5}$$

Both (3.4) and (3.5) refer only to the spectral data $\{(\lambda_k, v_k)\}_k$, and neither makes essential use of the dimension d. In Section 4 we replace the eigenvectors v_k by the eigenfunctions ψ_k of T_R and the eigenvalues of Σ by those of T_R.

4 Stein's equation in infinite dimensions

Motivated by the spectral form (3.4), we now define Stein's equation on $C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$. Throughout, λ_k and ψ_k are the eigenvalues and eigenfunctions of the covariance operator T_R (Definition 2.4), and G is the Gaussian tangent field (Definition 2.6).

Definition 4.1 (Stein's equation)**.** *For $g, f \in C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ and a functional $\Lambda \colon C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}) \to \mathbb{R}$ with sufficiently many bounded Fréchet derivatives, define the spectral gradient and Laplacian terms*

$$(g \cdot \nabla \Lambda)(f) \coloneqq \sum_{k \geq 1} \langle g, \psi_k \rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})} \, (D\Lambda)(f)[\psi_k], \qquad (\Delta \Lambda)(f) \coloneqq \sum_{k \geq 1} \lambda_k \, (D^2 \Lambda)(f)[\psi_k, \psi_k].$$

Given $\Gamma \in \mathcal{F}$, we say that Λ solves the Stein equation *if*

$$(\Delta \Lambda - f \cdot \nabla \Lambda)(f) = \Gamma(f) - \mathbb{E}\, \Gamma(G), \qquad f \in C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}). \tag{4.1}$$

Equation (4.1) is the infinite-dimensional analogue of (3.4). The eigenvectors v_k are replaced by the eigenfunctions ψ_k, the directional derivatives by Fréchet derivatives along ψ_k, and the eigenvalues of Σ by those of T_R.

4.1 The candidate solution

Mirroring (3.5), we propose the candidate solution $\Lambda = U_o\Gamma$ defined, for $f \in C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$, by

$$U_o\Gamma(f) = \int_0^1 \frac{1}{2t}\Big[\,\mathbb{E}\,\Gamma\big(\sqrt{t}\,f + \sqrt{1-t}\,G\big) - \mathbb{E}\,\Gamma(G)\Big]\,dt. \tag{4.2}$$

Expanding f and G in the orthonormal basis $\{\psi_k\}_k$ via $f = \sum_k \langle f, \psi_k\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\psi_k$ and the Karhunen–Loève expansion (2.1), formula (4.2) coincides with the spectral expression

$$U_o\Gamma(f) = \int_0^1 \frac{1}{2t}\,\mathbb{E}\Big[\,\Gamma\Big(\textstyle\sum_k \sqrt{t}\,\langle f, \psi_k\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\psi_k + \sqrt{1-t}\,\lambda_k^{1/2}\eta_k\psi_k\Big) - \Gamma\Big(\textstyle\sum_k \lambda_k^{1/2}\eta_k\psi_k\Big)\,\Big]\,dt,$$

the direct analogue of (3.5).

We write the interpolation function used throughout this work. For $f \in C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ and $t \in [0,1]$, we define

$$Z_t(f) \coloneqq \sqrt{t}\,f + \sqrt{1-t}\,G, \tag{4.3}$$

so that $Z_0(f) = G$ and $Z_1(f) = f$.

We first show that $U_o\Gamma$ is well defined. The verification that $U_o\Gamma$ solves (4.1) is deferred to Section 5, since it requires a Hilbert-space integration-by-parts identity.

Lemma 4.2. *Let $\Gamma\colon C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}) \to \mathbb{R}$ be Fréchet differentiable with $\|D\Gamma\|_{\mathrm{op}} < \infty$. Then $|U_o\Gamma(f)| < \infty$ for every $f \in C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$.*

Proof. Fix $t \in (0,1)$ and, for $s \in [0,1]$, set $\gamma(s) = G + s\big(\sqrt{t}\,f + (\sqrt{1-t}-1)G\big)$. By the fundamental theorem of calculus and the bound $\|D\Gamma\|_{\mathrm{op}} < \infty$,

$$\begin{aligned}
\big|\Gamma(\sqrt{t}\,f + \sqrt{1-t}\,G) - \Gamma(G)\big| &\le \int_0^1 \big|D\Gamma(\gamma(s))[\sqrt{t}\,f + (\sqrt{1-t}-1)G]\big|\,ds \\
&\le \|D\Gamma\|_{\mathrm{op}} \int_0^1 \|\sqrt{t}\,f + (\sqrt{1-t}-1)G\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\,ds \\
&\le \sqrt{t}\,\|D\Gamma\|_{\mathrm{op}}\,\|f\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} + t\,\|D\Gamma\|_{\mathrm{op}}\,\|G\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})},
\end{aligned}$$

where we used that $|\sqrt{1-t}-1| \le t$ for $t \in (0,1)$, that the operator norm of $D\Gamma$ refers to the $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ norm (Definition 2.9), and that f and (a.s.) G are continuous on the compact set $\mathcal{S}_{\bar{\mu}}\mathcal{M}$ (Remark 2.1), hence in $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$. Inserting this into (4.2),

$$\begin{aligned}
|U_o\Gamma(f)| &\le \int_0^1 \frac{1}{2t}\,\mathbb{E}\Big[\sqrt{t}\,\|D\Gamma\|_{\mathrm{op}}\|f\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} + t\,\|D\Gamma\|_{\mathrm{op}}\|G\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\Big]\,dt \\
&= \|D\Gamma\|_{\mathrm{op}}\|f\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} + \tfrac{1}{2}\,\|D\Gamma\|_{\mathrm{op}}\,\mathbb{E}\|G\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} < \infty.
\end{aligned}$$

Since $\mathbb{E}\|G\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} \le \big(\mathbb{E}\|G\|^2_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\big)^{1/2} = \big(\sum_{k\ge1}\lambda_k\big)^{1/2} < \infty$ by (2.1) and Lemma 2.5. □

5 Hilbert–Gaussian integration by parts

The verification that $U_o\Gamma$ solves (4.1) rests on a Gaussian integration-by-parts formula valid on the Hilbert space $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$. We first state the Cameron–Martin theorem in the form we need here, and then prove the integration-by-parts identity.

5.1 The Cameron–Martin theorem on the cone

Let γ be the centered Gaussian measure on $E = L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ with covariance operator T_R and Cameron–Martin space H (Section 2.3).

Proposition 5.1 (Cameron–Martin, [6, Ch. 2])**.** *For every direction $h \in H$, the shifted measure $\gamma_h(\cdot) = \gamma(\cdot - h)$ is absolutely continuous with respect to γ, with Radon–Nikodym derivative*

$$\frac{d\gamma_h}{d\gamma}(x) = \exp\Big\{ \langle T_R^{-1/2} x,\, T_R^{-1/2} h\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} - \tfrac{1}{2}\|T_R^{-1/2} h\|^2_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} \Big\}, \qquad x \in E. \tag{5.1}$$

Consequently, for any functional $F\colon E \to \mathbb{R}$ and $X \sim \gamma$,

$$\mathbb{E}\big[F(X+h)\big] = \mathbb{E}\Big[F(X)\, \exp\Big\{ \langle X,\, T_R^{-1} h\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} - \tfrac{1}{2}\langle T_R^{-1/2} h,\, T_R^{-1/2} h\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} \Big\}\Big]. \tag{5.2}$$

5.2 The integration-by-parts formula

Proposition 5.2 (Hilbert–Gaussian integration by parts)**.** *Let γ be the centered Gaussian measure on $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ with covariance operator T_R, with eigenpairs (λ_k, ψ_k). Let $F\colon C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}) \to \mathbb{R}$ be Fréchet differentiable with $\sup_g \|DF(g)\|_{\mathrm{op}} < \infty$. Then, for $X \sim \gamma$ and every k with $\lambda_k > 0$,*

$$\mathbb{E}\big(\langle X, \psi_k\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\, F(X)\big) = \lambda_k\, \mathbb{E}\big(DF(X)[\psi_k]\big). \tag{5.3}$$

Remark 5.3. Identity (5.3) is the exact analogue of the finite-dimensional fact that, for a centered Gaussian vector $X = (X_1, \dots, X_n)$ with diagonal covariance, $\mathbb{E}[X_k F(X)] = \lambda_k\, \mathbb{E}[\partial_k F(X)]$. The diagonalization is performed here by the spectral decomposition of T_R, the eigenfunctions ψ_k playing the role of the coordinate axes.

Proof. Let $h \in H$ and consider $\mathbb{E}[F(X + th)]$ for $t > 0$. Differentiating at $t = 0$, the left-hand side gives, by the chain rule and the boundedness of DF (which permits differentiation under the expectation, see [10]),

$$\frac{d}{dt}\, \mathbb{E}\big[F(X+th)\big]\Big|_{t=0} = \mathbb{E}\big(DF(X)[h]\big).$$

Differentiating the right-hand side of (5.2) at $t = 0$ yields $\mathbb{E}\big(F(X)\, \langle X, T_R^{-1} h\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\big)$. Equating,

$$\mathbb{E}\big(DF(X)[h]\big) = \mathbb{E}\big(\langle X, T_R^{-1} h\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\, F(X)\big).$$

For $\lambda_k > 0$ we have $T_R^{-1}\psi_k = \lambda_k^{-1}\psi_k$, and $\psi_k \in H$; taking $h = \psi_k$ gives $\mathbb{E}(DF(X)[\psi_k]) = \lambda_k^{-1}\, \mathbb{E}\big(\langle X, \psi_k\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} F(X)\big)$, which is (5.3). □

A direct consequence of the previous result lowers a second Fréchet derivative to a first one against a Gaussian coordinate.

Lemma 5.4. *Fix $h \in C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ and $t \in (0,1)$. For $\Gamma \in \mathcal{F}$ and $G_k = \langle G, \psi_k\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}$,*

$$\mathbb{E}_G\Big(G_k\, D\Gamma\big(\sqrt{t}\, h + \sqrt{1-t}\, G\big)[\psi_k]\Big) = \sqrt{1-t}\, \lambda_k\, \mathbb{E}_G\Big(D^2\Gamma\big(\sqrt{t}\, h + \sqrt{1-t}\, G\big)[\psi_k, \psi_k]\Big).$$

Proof. Set $\zeta_t(G) = \sqrt{t}\, h + \sqrt{1-t}\, G$ and $F(G) = D\Gamma(\zeta_t(G))[\psi_k]$. Since $D\zeta_t(G)[\psi_k] = \sqrt{1-t}\, \psi_k$, the chain rule gives $DF(G)[\psi_k] = \sqrt{1-t}\, D^2\Gamma(\zeta_t(G))[\psi_k, \psi_k]$. Applying Proposition 5.2 to F,

$$\mathbb{E}_G\big(G_k F(G)\big) = \lambda_k\, \mathbb{E}_G\big(DF(G)[\psi_k]\big) = \sqrt{1-t}\, \lambda_k\, \mathbb{E}_G\big(D^2\Gamma(\zeta_t(G))[\psi_k, \psi_k]\big). \qquad \square$$

5.3 Verification of the solution

Theorem 5.5. *Let $\Gamma \in \mathcal{F}$. Then $\Lambda = U_o\Gamma$, given by* (4.2)*, solves the Stein equation* (4.1).

Proof. Fix $f \in C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ and write $f_k = \langle f, \psi_k\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}$. Expanding the interpolation (4.3) in the eigenbasis,

$$Z_t(f) = \sqrt{t}\sum_{k\geq 1} f_k\psi_k + \sqrt{1-t}\sum_{k\geq 1}\lambda_k^{1/2}\eta_k\psi_k,$$

so that $Z_0(f) = G$ and $Z_1(f) = f$. Since $\Gamma \in \mathcal{F}$ has uniformly bounded Fréchet derivatives, we may differentiate under the expectation [10] and apply the fundamental theorem of calculus to obtain

$$\begin{aligned}\Gamma(f) - \mathbb{E}\,\Gamma(G) &= \int_0^1 \frac{d}{dt}\,\mathbb{E}\,\Gamma(Z_t(f))\,dt = \int_0^1 \mathbb{E}\,D\Gamma(Z_t(f))\Big[\tfrac{1}{2\sqrt{t}}f - \tfrac{1}{2\sqrt{1-t}}G\Big]\,dt \\ &= \underbrace{\frac{1}{2}\int_0^1 \sum_k \frac{f_k}{\sqrt{t}}\,\mathbb{E}\,D\Gamma(Z_t(f))[\psi_k]\,dt}_{=:A} - \underbrace{\frac{1}{2}\int_0^1 \sum_k \mathbb{E}\,\frac{\lambda_k^{1/2}\eta_k}{\sqrt{1-t}}\,D\Gamma(Z_t(f))[\psi_k]\,dt}_{=:B}.\end{aligned} \tag{5.4}$$

We identify A and B with the two terms of (4.1).

For A, differentiating the integral form of $\Lambda = U_o\Gamma$ in the direction ψ_k,

$$\begin{aligned}(f\cdot\nabla\Lambda)(f) &= \sum_k f_k\, D\Lambda(f)[\psi_k] = \sum_k f_k \int_0^1 \frac{1}{2t}\,\mathbb{E}\,D\Gamma(Z_t(f))[\sqrt{t}\,\psi_k]\,dt \\ &= \frac{1}{2}\sum_k f_k\int_0^1 \frac{1}{\sqrt{t}}\,\mathbb{E}\,D\Gamma(Z_t(f))[\psi_k]\,dt = A.\end{aligned}$$

For B, differentiating twice and applying Lemma 5.4,

$$\begin{aligned}(\Delta\Lambda)(f) &= \sum_k \lambda_k\, D^2\Lambda(f)[\psi_k,\psi_k] = \sum_k \lambda_k\int_0^1 \frac{1}{2}\,\mathbb{E}\,D^2\Gamma(Z_t(f))[\psi_k,\psi_k]\,dt \\ &= \frac{1}{2}\int_0^1 \frac{1}{\sqrt{1-t}}\sum_k \mathbb{E}\big(\langle G,\psi_k\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\,D\Gamma(Z_t(f))[\psi_k]\big)\,dt = B.\end{aligned}$$

Substituting A and B into (5.4) gives $\Gamma(f) - \mathbb{E}\,\Gamma(G) = (f\cdot\nabla\Lambda)(f) - (\Delta\Lambda)(f)$, which is exactly (4.1). The interchange of the infinite sum, the t-integral and the expectation is justified at each step by the uniform bounds $\sup_g \|D^j\Gamma(g)\|_{\mathrm{op}} \leq 1$ ($j = 1, 2$) and the trace-class property $\sum_k \lambda_k < \infty$ (Lemma 2.5). □

6 A convergence rate in the smooth metric

We now use the solution of the Stein equation to bound the smooth distance $d_{\mathcal{F}}(\bar{\tau}_n, G)$. Recall the standardized empirical field (Eq. 1.2) and define its leave-one-out variant,

$$\bar{\tau}_n = \frac{1}{\sqrt{n}}\sum_{i=1}^n \tau_i, \qquad \bar{\tau}_n^{(j)} = \frac{1}{\sqrt{n}}\sum_{\substack{i=1\\ i\neq j}}^n \tau_i,$$

so that $\bar{\tau}_n - \bar{\tau}_n^{(i)} = \tau_i/\sqrt{n}$. The main result of this section is the following.

Theorem 6.1. *Let $\tau_1, \dots, \tau_n$ be i.i.d. induced tangent fields on $\mathcal{S}_{\bar{\mu}}\mathcal{M}$ with bounded third moment, i.e., $\mathbb{E}\|\tau\|^3_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} < \infty$, and let G be the Gaussian tangent field with covariance R. Then for every $\Gamma \in \mathcal{F}$,*

$$\mathbb{E}\,\Gamma(\bar{\tau}_n) - \mathbb{E}\,\Gamma(G) = \mathbb{E}\big[(\bar{\tau}_n \cdot \nabla\Lambda)(\bar{\tau}_n) - (\Delta\Lambda)(\bar{\tau}_n)\big] = \mathcal{O}(n^{-1/2}), \tag{6.1}$$

where $\Lambda = U_o\Gamma$. In particular $d_{\mathcal{F}}(\bar{\tau}_n, G) = \mathcal{O}(n^{-1/2})$. The constant implicit in $\mathcal{O}(n^{-1/2})$ depends only on $\mathbb{E}\|\tau\|^3_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}$ and $\mathrm{Tr}(T_R)$. In particular. it is uniform over $\Gamma \in \mathcal{F}$, a fact used in Section 7.

The strategy is a leave-one-out Taylor expansion of Λ, after which the error reduces to a third-derivative remainder and a stability term, each of order $\mathcal{O}(1)$ in n.

6.1 The error decomposition

Since $\{\psi_k\}_k$ is an orthonormal basis and $D\Lambda$ is uniformly bounded (by Lemma 6.2 below and $\Gamma \in \mathcal{F}$),

$$(D\Lambda)(\bar{\tau}_n)[\tau_i] = \sum_{k\geq 1} \langle \tau_i, \psi_k\rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\,(D\Lambda)(\bar{\tau}_n)[\psi_k].$$

Apply Taylor's theorem [5, Theorem 12.4.4] with $m = 2$ to $F(\cdot) = D\Lambda(\cdot)[\tau_i]$ at $u = \bar{\tau}_n^{(i)}$ with increment $h = \tau_i/\sqrt{n}$:

$$D\Lambda(\bar{\tau}_n)[\tau_i] = D\Lambda(\bar{\tau}_n^{(i)})[\tau_i] + \frac{1}{\sqrt{n}}\,D^2\Lambda(\bar{\tau}_n^{(i)})[\tau_i,\tau_i] + R_2, \tag{6.2}$$

with integral remainder

$$R_2 = \frac{1}{n}\int_0^1 (1-s)\,D^3\Lambda\Big(\bar{\tau}_n^{(i)} + s\,\tfrac{\tau_i}{\sqrt{n}}\Big)[\tau_i,\tau_i,\tau_i]\,ds. \tag{6.3}$$

Expanding τ_i in the eigenbasis and taking expectations, the centered and diagonal structure of Lemma 2.7 eliminates the first-order term and diagonalizes the second:

$$\mathbb{E}\,(D\Lambda)(\bar{\tau}_n)[\tau_i] = \frac{1}{\sqrt{n}}\sum_{k\geq 1}\lambda_k\,\mathbb{E}\,(D^2\Lambda)(\bar{\tau}_n)[\psi_k,\psi_k] + \frac{1}{n}\,\widetilde{E}_i(\bar{\tau}_n^{(i)}), \tag{6.4}$$

where

$$\widetilde{E}_i(\bar{\tau}_n^{(i)}) = \mathrm{Err}_n + n\,\mathbb{E}[R_2], \qquad \mathrm{Err}_n = \sqrt{n}\sum_{k\geq 1}\lambda_k\Big(\mathbb{E}(D^2\Lambda)(\bar{\tau}_n^{(i)})[\psi_k,\psi_k] - \mathbb{E}(D^2\Lambda)(\bar{\tau}_n)[\psi_k,\psi_k]\Big). \tag{6.5}$$

Multiplying (6.4) by $n^{-1/2}$, summing over $i \in [n]$, and using Definition 4.1 together with Theorem 5.5, we obtain

$$\mathbb{E}\,\Gamma(\bar{\tau}_n) - \mathbb{E}\,\Gamma(G) = \mathbb{E}\big[(\bar{\tau}_n \cdot \nabla\Lambda)(\bar{\tau}_n) - (\Delta\Lambda)(\bar{\tau}_n)\big] = n^{-3/2}\sum_{i=1}^{n}\widetilde{E}_i(\bar{\tau}_n^{(i)}). \tag{6.6}$$

Theorem 6.1 therefore follows once we show $\widetilde{E}_i(\bar{\tau}_n^{(i)}) = \mathcal{O}(1)$ in n, which we do by bounding the two terms in (6.5) separately. We use the following differentiation lemma, whose proof is a routine application of dominated convergence and the bound $\sup_g \|D^j\Gamma(g)\|_{\mathrm{op}} \leq 1$ [10].

Lemma 6.2. *For $\Gamma \in \mathcal{F}$, the functional $\Lambda = U_o\Gamma$ is three times Fréchet differentiable, the derivatives may be computed by differentiating under the integral sign in* (4.2)*, and for $j = 1, 2, 3$,*

$$D^j\Lambda(h)[\psi_{k_1},\dots,\psi_{k_j}] = \int_0^1 \frac{t^{j/2}}{2t}\,\mathbb{E}_G\,D^j\Gamma(Z_t(h))[\psi_{k_1},\dots,\psi_{k_j}]\,dt,$$

so that in particular $\sup_h \|D^j\Lambda(h)\|_{\mathrm{op}} < \infty$.

6.2 Bounding the stability term

Lemma 6.3. *Under the hypotheses of Theorem 6.1,* $\mathrm{Err}_n = \mathcal{O}(1)$.

Proof. By Lemma 6.2, for $h \in C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$,

$$D^2\Lambda(h)[\psi_k,\psi_k] = \frac{1}{2}\int_0^1 \mathbb{E}_G\, D^2\Gamma(Z_t(h))[\psi_k,\psi_k]\,dt,$$

so the difference $\big|D^2\Lambda(\bar{\tau}_n^{(i)})[\psi_k,\psi_k] - D^2\Lambda(\bar{\tau}_n)[\psi_k,\psi_k]\big|$ is bounded by

$$\frac{1}{2}\int_0^1 \mathbb{E}_G\big|D^2\Gamma(Z_t(\bar{\tau}_n^{(i)}))[\psi_k,\psi_k] - D^2\Gamma(Z_t(\bar{\tau}_n))[\psi_k,\psi_k]\big|\,dt.$$

Since $Z_t(\bar{\tau}_n) = Z_t(\bar{\tau}_n^{(i)}) + \sqrt{t/n}\,\tau_i$ and the third Fréchet derivative of $\Gamma \in \mathcal{F}$ has operator norm at most 1, the mean value theorem $D^2\Gamma(a)[\psi_k,\psi_k] - D^2\Gamma(b)[\psi_k,\psi_k] = \int_0^1 D^3\Gamma(b + s(a-b))[a-b,\psi_k,\psi_k]\,ds$ gives, with c the universal constant of the polarization inequality $|A[g_1,g_2,g_3]| \le c\|A\|_{\mathrm{op}}\prod_i \|g_i\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}$ for symmetric trilinear forms (recall that $\|\cdot\|_{\mathrm{op}}$ in Definition 2.9 is a supremum over the diagonal),

$$\begin{aligned}&\Big|D^2\Gamma(Z_t(\bar{\tau}_n^{(i)}))[\psi_k,\psi_k] - D^2\Gamma(Z_t(\bar{\tau}_n))[\psi_k,\psi_k]\Big| \\ &\qquad \le c\,\big\|Z_t(\bar{\tau}_n^{(i)}) - Z_t(\bar{\tau}_n)\big\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} \le c\,\frac{\sqrt{t}}{\sqrt{n}}\,\|\tau_i\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}.\end{aligned}$$

Hence $\big|D^2\Lambda(\bar{\tau}_n^{(i)})[\psi_k,\psi_k] - D^2\Lambda(\bar{\tau}_n)[\psi_k,\psi_k]\big| \le C\,n^{-1/2}\|\tau_i\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}$, and therefore, since Err_n is defined through expectations,

$$|\mathrm{Err}_n| \le \sqrt{n}\sum_{k\ge1}\lambda_k\,\frac{C\,\mathbb{E}\|\tau_i\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}}{\sqrt{n}} = C\,\mathbb{E}\|\tau_i\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\sum_{k\ge1}\lambda_k < \infty,$$

the series converging since T_R is trace class (Lemma 2.5), and the expectation being finite by the third-moment hypothesis and Jensen's inequality. □

6.3 Bounding the remainder term

Lemma 6.4. *Under the hypotheses of Theorem 6.1,* $n\,\mathbb{E}[R_2] = \mathcal{O}(1)$.

Proof. Following [3, p. 23], set $h_s = \bar{\tau}_n^{(i)} + s\,\tau_i/\sqrt{n}$ and $Z_t(h_s) = \sqrt{t}\,h_s + \sqrt{1-t}\,G$. By Lemma 6.2,

$$\begin{aligned}n\,\mathbb{E}[R_2] &= \mathbb{E}\int_0^1 (1-s)\,D^3\Lambda(h_s)[\tau_i,\tau_i,\tau_i]\,ds \\ &= \mathbb{E}\int_0^1 (1-s)\int_0^1 \frac{\sqrt{t}}{2}\,\mathbb{E}_G\,D^3\Gamma(Z_t(h_s))[\tau_i,\tau_i,\tau_i]\,dt\,ds.\end{aligned}$$

Taking absolute values and using $\|D^3\Gamma\|_{\mathrm{op}} \le 1$,

$$|n\,\mathbb{E}[R_2]| \le \frac{1}{2}\,\mathbb{E}\|\tau_i\|^3_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})}\int_0^1\int_0^1 (1-s)\sqrt{t}\,ds\,dt \lesssim \mathbb{E}\|\tau_i\|^3_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})} = \mathcal{O}(1),$$

which is independent of n. □

Proof of Theorem 6.1. By (6.6), $\mathbb{E}\,\Gamma(\bar{\tau}_n) - \mathbb{E}\,\Gamma(G) = n^{-3/2}\sum_{i=1}^n \widetilde{E}_i(\bar{\tau}_n^{(i)})$, and by (6.5) together with Lemmas 6.3 and 6.4, each $\widetilde{E}_i(\bar{\tau}_n^{(i)}) = \mathrm{Err}_n + n\,\mathbb{E}[R_2] = \mathcal{O}(1)$. Summing the n identical-in-law terms,

$$\big|\mathbb{E}\,\Gamma(\bar{\tau}_n) - \mathbb{E}\,\Gamma(G)\big| \le n^{-3/2}\cdot n\cdot\mathcal{O}(1) = \mathcal{O}(n^{-1/2}).$$

Taking the supremum over $\Gamma \in \mathcal{F}$ yields $d_{\mathcal{F}}(\bar{\tau}_n, G) = \mathcal{O}(n^{-1/2})$. □

7 From the smooth metric to Wasserstein

The smooth metric $d_{\mathcal{F}}$ yields the fast rate (6.1), but it is unclear if it can imply tightness (see [1, Section 1.1]). To reach the Wasserstein distance we use a Gaussian smoothing argument, in the spirit of the technique developed for the Euclidean sphere by Balasubramanian, Goldstein, Ross and Salim [1]. Our implementation is finite-dimensional, the fields are truncated to the eigenmodes of T_R whose eigenvalues exceed a threshold t, and the test functional is mollified by a Gaussian on the finite-dimensional space of retained modes. Every estimate is derived from the spectrum of T_R; no geometric information on $\mathcal{S}_{\bar{\mu}}\mathcal{M}$ beyond its compactness is used. We keep the notation λ_k, ψ_k for the spectrum of T_R, and write $f_k = \langle f, \psi_k \rangle_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})}$ for the coefficients of $f \in L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$.

7.1 The spectral assumption and the spectral truncation

We work under the following decay and boundedness assumption on the spectrum.

Assumption 7.1 (Spectral assumption)**.** *The spectral decomposition of T_R, with the positive eigenvalues $\lambda_{(1)} \geq \lambda_{(2)} \geq \cdots > 0$ enumerated in nonincreasing order and with multiplicity (Remark 2.8), satisfies:*

(1) there exist $s > 1$ and $C_\lambda > 0$ such that $\lambda_{(j)} \leq C_\lambda\, j^{-2s}$ for all $j \geq 1$.
(2) the eigenfunctions with positive eigenvalue are uniformly bounded: there is $M > 0$ with $|\psi_k(V)| \leq M$ for all $k \geq 1$ with $\lambda_k > 0$ and $V \in \mathcal{S}_{\bar{\mu}}\mathcal{M}$.

Remark 7.2. Only an *upper* bound on the eigenvalues is required; in particular, vanishing eigenvalues are allowed (this happens for the kale, Section 8). The case $s = 1$ is excluded because it is exactly the borderline at which the estimate of Lemma 7.3 fails.

For a threshold $t \in (0, 1]$ we set

$$K_t \coloneqq \{k \geq 1 : \lambda_k \geq t\}, \quad N_t \coloneqq \#K_t, \quad P_t f \coloneqq \sum_{k \in K_t} f_k\, \psi_k, \quad \pi_t f \coloneqq (f_k)_{k \in K_t} \in \mathbb{R}^{N_t}, \tag{7.1}$$

so that P_t is the orthogonal projection of $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$ onto the finite-dimensional space $V_t \coloneqq \operatorname{span}\{\psi_k : k \in K_t\}$, and $\pi_t \colon L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}) \to \mathbb{R}^{N_t}$ is a linear contraction ($|\pi_t f| \leq \|f\|_{L^2}$, with $|\cdot|$ the Euclidean norm). Note that, since $N_t = \#\{j : \lambda_{(j)} \geq t\}$ and $\lambda_{(j)} \geq t$ forces $C_\lambda\, j^{-2s} \geq t$ by Assumption 7.1(1),

$$N_t \leq \Big(\frac{C_\lambda}{t}\Big)^{\frac{1}{2s}}. \tag{7.2}$$

The *spectral truncation* $P_t F$ of a random field F with continuous paths is defined pathwise; it is a random element of $V_t \subset C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R})$.

The kale example of Section 8 satisfies Assumption 7.1 with $s = 3$.

Lemma 7.3 (Truncation error)**.** *Suppose T_R satisfies Assumption 7.1, and let F be either the Gaussian tangent field G or the standardized empirical field $\bar{\tau}_n$ for some $n \geq 1$. Then for every $t \in (0, 1]$,*

$$\mathbb{E}\|F - P_t F\|_\infty \leq M \sum_{k \colon \lambda_k < t} \sqrt{\lambda_k} \leq M\Big(2 + \frac{C_\lambda^{1/2}}{s-1}\Big)\, t^{\frac{s-1}{2s}}. \tag{7.3}$$

In particular the bound does not depend on n.

Proof. Write $F_k = \langle F, \psi_k \rangle_{L^2}$. We first claim that $\mathbb{E}|F_k| \leq \sqrt{\lambda_k}$ for every k. For $F = G$ this is immediate from $G_k = \sqrt{\lambda_k}\, \eta_k$ in (2.1). For $F = \bar{\tau}_n$, Lemma 2.7 and the independence

of the τ_i give $\mathbb{E}\langle\bar{\tau}_n,\psi_k\rangle_{L^2}^2 = \frac{1}{n}\sum_{i=1}^n \mathbb{E}\langle\tau_i,\psi_k\rangle_{L^2}^2 = \lambda_k$, and the claim follows from the Cauchy–Schwarz inequality. In particular $F_k = 0$ almost surely whenever $\lambda_k = 0$, so that, almost surely, $F = \sum_{k:\lambda_k>0} F_k\psi_k$ in $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$ (Remark 2.8).

Now, observe that $\mathbb{E}\sum_{k:\lambda_k>0}|F_k|\,\|\psi_k\|_\infty \le M\sum_k\sqrt{\lambda_k} < \infty$ (which converges by Assumption 7.1(1), since $s>1$), so almost surely the series $\sum_{k:\lambda_k>0}F_k\psi_k$ converges absolutely and uniformly on $\mathcal{S}_{\bar{\mu}}\mathcal{M}$. Its sum is a continuous function which coincides with F in $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$, hence almost everywhere; since F has continuous sample paths and the reference measure on $\mathcal{S}_{\bar{\mu}}\mathcal{M}$ charges every nonempty open set (Remark 2.1), the two functions coincide everywhere. Therefore $F - P_tF = \sum_{k:\,0<\lambda_k<t}F_k\psi_k$ uniformly, consequently we have $\|F-P_tF\|_\infty \le M\sum_{k:\lambda_k<t}|F_k|$, and taking expectations gives the first inequality in (7.3).

For the second inequality, observe that zero eigenvalues do not contribute to the sum in (7.3), and that its value does not depend on the order of summation, so it may be computed along the nonincreasing enumeration $(\lambda_{(j)})_j$ of the positive eigenvalues. Then put $K = \lceil t^{-1/(2s)}\rceil$ which we use to split the sum below, so that $t^{-1/(2s)} \le K \le 2t^{-1/(2s)}$. Thus, splitting the sum and using that $\sqrt{\lambda_{(j)}} \le C_\lambda^{1/2} j^{-s}$,

$$\begin{aligned}\sum_{k:\,\lambda_k<t}\sqrt{\lambda_k} &= \sum_{j:\,\lambda_{(j)}<t}\sqrt{\lambda_{(j)}} \le K\sqrt{t} + C_\lambda^{1/2}\sum_{j>K} j^{-s} \le K\sqrt{t} + \frac{C_\lambda^{1/2}}{s-1}\,K^{1-s}\\ &\le 2\,t^{\frac12-\frac{1}{2s}} + \frac{C_\lambda^{1/2}}{s-1}\,t^{\frac{s-1}{2s}},\end{aligned}$$

which is (7.3). □

7.2 A smoothing inequality on the cone

Fix $t\in(0,1]$ and let $\eta = (\eta_k)_{k\in K_t}$ be i.i.d. standard Gaussian variables independent any other random element, and define

$$S_t \coloneqq \sum_{k\in K_t}\eta_k\,\psi_k \in V_t.$$

For $\zeta\in\mathcal{W}$ and $\delta\in(0,1]$ we define the *smoothed functional*

$$\zeta_{t,\delta}(f) \coloneqq \mathbb{E}\,\zeta\big(P_tf + \delta S_t\big), \qquad f\in C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R}). \tag{7.4}$$

Since ζ is evaluated only on $V_t\cong\mathbb{R}^{N_t}$, the smoothed functional is a Gaussian mollification in $\mathbb{R}^{N_t}$. Let

$$\psi_\zeta(x) \coloneqq \zeta\Big(\sum_{k\in K_t}x_k\psi_k\Big), \qquad x\in\mathbb{R}^{N_t}.$$

One has with

$$\varphi_\delta(z) \coloneqq (2\pi\delta^2)^{-N_t/2}e^{-|z|^2/(2\delta^2)},$$

that $\zeta_{t,\delta} = \Phi\circ\pi_t$ with $\Phi \coloneqq \psi_\zeta * \varphi_\delta$. Let He_j denote the probabilists' Hermite polynomials ($\mathrm{He}_0=1$, $\mathrm{He}_1(x)=x$, $\mathrm{He}_2(x)=x^2-1$) and $c_j \coloneqq \mathbb{E}|\mathrm{He}_j(Z)|$ for $Z\sim\mathcal{N}(0,1)$.

Lemma 7.4 (Mollification)**.** *Suppose T_R satisfies Assumption 7.1, and let $\zeta\in\mathcal{W}$, $t,\delta\in(0,1]$. Then:*

- *(a) $|\psi_\zeta(x)-\psi_\zeta(y)| \le M\sqrt{N_t}\,|x-y|$ for all $x,y\in\mathbb{R}^{N_t}$;*
- *(b) $\zeta_{t,\delta}$ is three times Fréchet differentiable on $C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$ (with respect to the $L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$ norm, as in Section 2.4), and*

$$\sup_f\|D^{j+1}\zeta_{t,\delta}(f)\|_{\mathrm{op}} \le c_j\,M\sqrt{N_t}\,\delta^{-j}, \qquad j=0,1,2; \tag{7.5}$$

(c) for every random element F of $C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$ independent of η,

$$\big|\mathbb{E}\,\zeta(F)-\mathbb{E}\,\zeta_{t,\delta}(F)\big|\le\mathbb{E}\|F-P_tF\|_\infty+\delta\,M\,N_t. \tag{7.6}$$

Proof. (a) Since ζ is 1-Lipschitz for $\|\cdot\|_\infty$ and $\|\psi_k\|_\infty\le M$, using the Cauchy–Schwarz inequality $|\psi_\zeta(x)-\psi_\zeta(y)|\le\big\|\sum_{k\in K_t}(x_k-y_k)\psi_k\big\|_\infty\le M\sum_{k\in K_t}|x_k-y_k|\le M\sqrt{N_t}\,|x-y|$.

(b) Let $u\in\mathbb{R}^{N_t}$ be a unit vector and $j\ge0$. Splitting $z=\langle z,u\rangle u+z^\perp$, the density φ_δ factorizes, and Rodrigues' formula for the one-dimensional Gaussian density gives

$$\partial_u^j\varphi_\delta(z)=(-1)^j\delta^{-j}\,\mathrm{He}_j\big(\langle z,u\rangle/\delta\big)\,\varphi_\delta(z)\qquad\Longrightarrow\qquad\int_{\mathbb{R}^{N_t}}|\partial_u^j\varphi_\delta(z)|\,dz=c_j\,\delta^{-j}.$$

Since ψ_ζ is Lipschitz, $\Phi=\psi_\zeta*\varphi_\delta$ is C^∞ with $\partial_u^j\Phi=\psi_\zeta*\partial_u^j\varphi_\delta$, and by (a)

$$|\partial_u^j\Phi(x)-\partial_u^j\Phi(y)|\le\int|\psi_\zeta(x-z)-\psi_\zeta(y-z)|\,|\partial_u^j\varphi_\delta(z)|\,dz\le c_j\,M\sqrt{N_t}\,\delta^{-j}\,|x-y|. \tag{7.7}$$

In particular all derivatives of Φ of order at least 1 are bounded. Since π_t is a linear contraction from $(C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R}),\|\cdot\|_{L^2})$ to $\mathbb{R}^{N_t}$, the chain rule shows that $\zeta_{t,\delta}=\Phi\circ\pi_t$ is Fréchet-C^∞ with $D^j\zeta_{t,\delta}(f)[g,\dots,g]=\partial_v^j\Phi(\pi_tf)$, with $v=\pi_tg$. Writing $v=|v|u$ with u a unit vector, $\partial_v^j=|v|^j\partial_u^j$, and (7.7) gives

$$\begin{aligned}\big|D^{j+1}\zeta_{t,\delta}(f)[g,\dots,g]\big|&=\lim_{r\to0}\frac{\big|\partial_v^j\Phi(\pi_tf+rv)-\partial_v^j\Phi(\pi_tf)\big|}{r}\\&\le c_j\,M\sqrt{N_t}\,\delta^{-j}\,|v|^{j+1}\le c_j\,M\sqrt{N_t}\,\delta^{-j}\,\|g\|_{L^2}^{j+1},\end{aligned}$$

which is (7.5) by Definition 2.9.

(c) Since ζ is 1-Lipschitz for $\|\cdot\|_\infty$ and F is independent of η, $|\mathbb{E}\,\zeta(F)-\mathbb{E}\,\zeta(P_tF+\delta S_t)|\le\mathbb{E}\|F-P_tF\|_\infty+\delta\,\mathbb{E}\|S_t\|_\infty$, and $\mathbb{E}\|S_t\|_\infty\le M\sum_{k\in K_t}\mathbb{E}|\eta_k|\le MN_t$. □

Proposition 7.5 (Smoothing inequality)**.** *Suppose T_R satisfies Assumption 7.1, and let M be the bound of Assumption 7.1(2). Then for every $\zeta\in\mathcal{W}$ and $t,\delta\in(0,1]$, the functional $\big(M\sqrt{N_t}\,\delta^{-2}\big)^{-1}\zeta_{t,\delta}$ belongs to $\mathcal{F}$. Consequently, for all random elements F,F' of $C(\mathcal{S}_{\bar{\mu}}\mathcal{M},\mathbb{R})$,*

$$d_{\mathcal{W}}(F,F')\le M\sqrt{N_t}\,\delta^{-2}\,d_{\mathcal{F}}(F,F')+\mathbb{E}\|F-P_tF\|_\infty+\mathbb{E}\|F'-P_tF'\|_\infty+2\,\delta\,M\,N_t. \tag{7.8}$$

Proof. By (7.5) and $\delta\le1$, the three derivatives of $\zeta_{t,\delta}$ have operator norm at most $\max\{c_0,c_1,c_2\}\,M\sqrt{N_t}\,\delta^{-2}$, where c_0,c_1,c_2 are the constants of Lemma 7.4(b). Hence, $(M\sqrt{N_t}\,\delta^{-2})^{-1}\zeta_{t,\delta}\in\mathcal{F}$ by Definition 2.10, and by the definition (1.4) of $d_{\mathcal{F}}$,

$$\big|\mathbb{E}\,\zeta_{t,\delta}(F)-\mathbb{E}\,\zeta_{t,\delta}(F')\big|\le M\sqrt{N_t}\,\delta^{-2}\,d_{\mathcal{F}}(F,F').$$

Adding (7.6) for F and for F' and taking the supremum over $\zeta\in\mathcal{W}$ yields (7.8). □

7.3 Proof of the main theorem

Proof of Theorem 1.1. Apply Proposition 7.5 with $F=\bar{\tau}_n$ and $F'=G$. The first term of (7.8) is bounded by Theorem 6.1 together with (7.2), the second and third by Lemma 7.3, and the last by (7.2) again. Hence, for all $t,\delta\in(0,1]$,

$$d_{\mathcal{W}}(\bar{\tau}_n,G)\le C\Big(\delta^{-2}\,t^{-\frac{1}{4s}}\,n^{-1/2}+t^{\frac{s-1}{2s}}+\delta\,t^{-\frac{1}{2s}}\Big), \tag{7.9}$$

with C independent of n,t,δ. Put $\nu:=\frac{1}{2s}\in(0,\frac12)$ and choose $\delta=t^{1/2}$: the last two terms are then both equal to $t^{\frac12-\nu}$, and the first becomes $t^{-1-\nu/2}n^{-1/2}$. Choosing $t=n^{-1/(3-\nu)}$ makes the first term equal to $t^{\frac12-\nu}$ as well, so that

$$d_{\mathcal{W}}(\bar{\tau}_n,G)\le3C\,t^{\frac12-\nu}=3C\,n^{-\frac{1/2-\nu}{3-\nu}}=3C\,n^{-\frac{s-1}{6s-1}},$$

which is (1.3) and completes the proof. □

Corollary 1.3 follows since convergence in $d_{\mathcal{W}}$ implies weak convergence on the Polish space $(C(\mathcal{S}_{\bar{\mu}}\mathcal{M}, \mathbb{R}), \|\cdot\|_\infty)$, as recalled in Definition 2.11.

8 An example: the kale

We illustrate the construction on the *kale*, the planar cone with an isolated hyperbolic singularity studied in [9], and verify that Assumption 7.1 holds with $s = 3$. Following [9], the kale is

$$\mathcal{K} = \big((0,\infty) \times (\mathbb{R}/\alpha\mathbb{Z})\big) \cup \{\mathbf{0}\},$$

where $\alpha > 2\pi$ is the total angle at the isolated apex $\mathbf{0}$, called the origin; the metric is not locally Euclidean there. Points are written in polar coordinates $p = (r, \theta)$ with $r > 0$ and $\theta \in \mathbb{R}/\alpha\mathbb{Z}$, and the origin is $\mathbf{0} = (0, \theta)$ for any θ. The set of directions at the origin is

$$\mathcal{S}_{\mathbf{0}}\mathcal{K} = \{(1, \theta) : \theta \in \mathbb{R}/\alpha\mathbb{Z}\},$$

a circle of length α; whenever a representative $\theta \in [0, \alpha]$ is used, the endpoints 0 and α are identified. Figure 1 shows the kale embedded in $\mathbb{R}^3$ as a union of seven equilateral triangles, for which $\alpha = 7\pi/3$; we fix this value of α from now on.

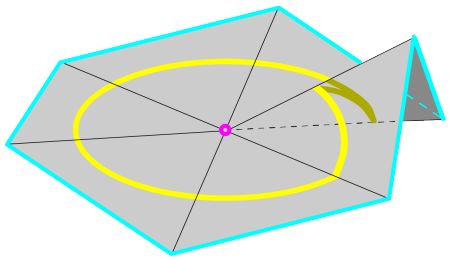


Figure 1: The kale, a two-dimensional object embedded in $\mathbb{R}^3$ as a union of seven equilateral triangles, so that the total angle at the origin is $\alpha = 7\pi/3$. The unit tangent cone $\mathcal{S}_{\mathbf{0}}\mathcal{K}$ is shown in yellow.

For $V, W \in \mathcal{S}_{\mathbf{0}}\mathcal{K}$ let $\theta = d_s(V, W)$ be the angular distance. By the geometry of the cone, the cosine of the angle is the piecewise function

$$\operatorname{kos}(\theta) = \begin{cases} \cos(\theta), & \theta \in [0, \pi], \\ -1, & \theta \in [\pi, \pi + \pi/3], \\ \cos(\theta - \pi/3), & \theta \in [\pi + \pi/3, 2\pi + \pi/3]. \end{cases} \tag{8.1}$$

The interval on which $\angle(W, V) = \pi$ has length $\pi/3$, and kos is continuous and α-periodic. Encoding $\log_{\mathbf{0}} x = (1, \theta_x)$ and $V = (1, \theta_V)$, the angular pairing (Definition 2.2) is $\langle \log_{\mathbf{0}} x, V\rangle_{\mathbf{0}} = \mathbf{d}(x, \mathbf{0}) \operatorname{kos}(\theta_x - \theta_V)$, and the induced tangent field is

$$\tau(x, V) = \mathbf{d}(x, \mathbf{0}) \operatorname{kos}(\theta_x - \theta_V) - \mathbb{E}_{x\sim\mu}\big[\mathbf{d}(x, \mathbf{0}) \operatorname{kos}(\theta_x - \theta_V)\big],$$

where $d_s(\log_{\mathbf{0}} x, V) = \theta_x - \theta_V \bmod (2\pi + \frac{\pi}{3})$. Integrating over x, the covariance is

$$\begin{aligned} R(V, W) = \mathbb{E}\big[\mathbf{d}(x, \mathbf{0})^2 \operatorname{kos}(\theta_x - \theta_V) \operatorname{kos}(\theta_x - \theta_W)\big] \\ - \mathbb{E}\big[\mathbf{d}(x, \mathbf{0}) \operatorname{kos}(\theta_x - \theta_V)\big]\, \mathbb{E}\big[\mathbf{d}(x, \mathbf{0}) \operatorname{kos}(\theta_x - \theta_W)\big], \end{aligned}$$

and the covariance operator is $T_R f(V) = \int_{\mathcal{S}_{\mathbf{0}}\mathcal{K}} R(V, W) f(W)\, dW$, where the measure on $\mathcal{S}_{\mathbf{0}}\mathcal{K}$ need only be finite and strictly positive [7, Theorem 2.4].

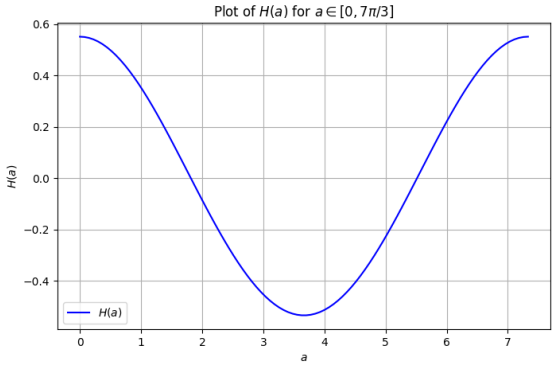


Figure 2: Numerical evaluation of $H(a)$ for $a \in [0, 7\pi/3]$.

Suppose μ is the uniform measure supported on $\mathcal{S}_{\mathbf{0}}\mathcal{K}$, so that sampling $x = (1, \theta) \sim \mu$ amounts to $\theta \sim \mathrm{unif}(0, 7\pi/3)$. Since $\mathcal{K}$ is $\mathrm{CAT}(0)$ and μ is invariant under the rotations $\theta \mapsto \theta + c$, which are isometries of $\mathcal{K}$, the origin $\mathbf{0}$ is the unique barycenter of μ. Integrating (8.1) over $[0, 7\pi/3]$ gives $\mathbb{E}_\mu[\mathrm{kos}] = -\frac{1}{7}$, so the centered field is

$$\tau(x, V) = \mathrm{kos}\big(d_s(\log_{\mathbf{0}} x, V)\big) + \tfrac{1}{7}, \qquad V \in \mathcal{S}_{\mathbf{0}}\mathcal{K},$$

and, with $a = \theta_V - \theta_W \bmod (7\pi/3)$ and $L = \frac{7\pi}{3}$,

$$R(V, W) = H(a) := \frac{1}{L}\int_0^L \mathrm{kos}(\phi)\,\mathrm{kos}(\phi + a)\,d\phi - \frac{1}{49}.$$

Writing kos as a Fourier series on $[0, L]$, $\mathrm{kos}(\phi) = \sum_{k\in\mathbb{Z}} c_k e^{2\pi i k\phi/L}$ with

$$c_k = \frac{1}{L}\int_0^L \mathrm{kos}(\phi)\, e^{-2\pi i k\phi/L}\, d\phi,$$

the autocorrelation diagonalizes and

$$H(a) = \sum_{k\in\mathbb{Z}} |c_k|^2 \cos\Big(\frac{2\pi k}{L}a\Big) - \frac{1}{49}. \tag{8.2}$$

Figure 2 shows $H(a)$ over $[0, 7\pi/3]$. The coefficients c_k are explicit, integrating (8.1) piece by piece gives

$$c_k = (-1)^k\,\frac{49\,\sin(\pi k/7)}{\pi k\,(36k^2 - 49)}, \qquad k \neq 0, \tag{8.3}$$

and $c_0 = -1/7$.

We equip $\mathcal{S}_{\mathbf{0}}\mathcal{K}$ with its arc-length measure $d\theta$. Since R is a translation-invariant kernel on a circle of circumference L, T_R is diagonalized by the trigonometric system, that is, by the real orthonormal basis

$$\psi_0 = \frac{1}{\sqrt{L}}, \qquad \psi_k^{\mathrm{c}}(\theta) = \sqrt{\tfrac{2}{L}}\,\cos\Big(\frac{2\pi k\theta}{L}\Big), \qquad \psi_k^{\mathrm{s}}(\theta) = \sqrt{\tfrac{2}{L}}\,\sin\Big(\frac{2\pi k\theta}{L}\Big), \qquad k \geq 1,$$

of $L^2(\mathcal{S}_{\mathbf{0}}\mathcal{K}, \mathbb{R})$. By (8.2), $T_R\psi_k^{\mathrm{c}} = \lambda_k\psi_k^{\mathrm{c}}$, $T_R\psi_k^{\mathrm{s}} = \lambda_k\psi_k^{\mathrm{s}}$ and $T_R\psi_0 = \lambda_0\psi_0$ with

$$\lambda_k = \begin{cases} L\,|c_k|^2, & k \geq 1, \\ L\big(|c_0|^2 - \frac{1}{49}\big) = 0, & k = 0. \end{cases} \tag{8.4}$$

In particular $\lambda_0 = 0$, and $\lambda_k = 0$ whenever 7 divides k. Note that $\ker T_R$ is therefore infinite-dimensional: it is the closed span of ψ_0 and of the functions $\psi_k^{\mathrm{c}}, \psi_k^{\mathrm{s}}$ with $k \geq 1$ divisible by 7.

We now verify Assumption 7.1 with $s = 3$. Each eigenvalue $\lambda_k = Lc_k^2$, $k \geq 1$, has multiplicity two (eigenfunctions $\psi_k^{\mathrm{c}}, \psi_k^{\mathrm{s}}$), and by (8.3),

$$\lambda_k = \frac{7 \cdot 49^2 \, \sin^2(\pi k/7)}{3\pi \, k^2 \, (36k^2 - 49)^2} \leq C_\lambda \, k^{-6}, \qquad C_\lambda := \frac{7 \cdot 49^2}{3\pi \cdot 13^2}, \qquad k \geq 1,$$

using $|36k^2 - 49| \geq 13k^2$. Hence, for every $t > 0$, the number of eigenvalues (counted with multiplicity) that are at least t is at most $2 \cdot \ |\{k \geq 1 : C_\lambda k^{-6} \geq t\}| \leq 2\,(C_\lambda/t)^{1/6}$, and taking $t = \lambda_{(j)}$, shows that the nonincreasing enumeration $\lambda_{(1)} \geq \lambda_{(2)} \geq \cdots$ of the positive eigenvalues (Remark 2.8) satisfies $\lambda_{(j)} \leq 2^6 C_\lambda \, j^{-6}$. Assumption 7.1(1) holds with $s = 3$ (and with the constant $2^6 C_\lambda$ in place of C_λ). Since $\sin^2(\pi k/7) \geq \sin^2(\pi/7)$ for $k \notin 7\mathbb{Z}$, the decay k^{-6} is attained, so the exponent $s = 3$ cannot be improved. The eigenfunctions are bounded by $\sqrt{2/L}$, so Assumption 7.1(2) holds with $M = \sqrt{2/L}$.

The remaining hypothesis of Theorem 1.1 is immediate. Since μ is supported on the unit sphere of the cone, $\mathbf{d}(x, \mathbf{0}) \equiv 1$ and $|\tau(x, V)| \leq |\operatorname{kos}(\cdot)| + \frac{1}{7} \leq 2$ for every x and V, so $\|\tau\|_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M})} \leq 2L^{1/2}$ almost surely and $\mathbb{E}\|\tau\|^3_{L^2(\mathcal{S}_{\bar{\mu}}\mathcal{M})} < \infty$.

Theorem 1.1 therefore applies to the kale with the uniform measure, with $s = 3$, and gives

$$d_{\mathcal{W}}(\bar{\tau}_n, G) = \mathcal{O}\big(n^{-2/17}\big).$$

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